YOUR 8-STEP JOURNEY TO AI MASTERY

AI & ML Foundations
Learn core concepts + cloud context

Cloud for AI
AWS, Azure, GCP essentials

Deep Learning & NLP
CNNs, RNNs, transformers, chatbots

Generative AI & LLMs
Prompting, prototyping, multimodal tools

GenAI Deployment
Docker, Kubernetes, scalable infra

MLOps & AIOps
Versioning, rollback, monitoring

Multi-Agent Systems
LangChain, n8n, memory-driven agents

Capstone Project
Build real, cloud-native AI solutions

✦ Role Evolution for Freshers & 1+ Years of Experience

Make the shift from surface learning to engineering production-grade GenAI systems, hands-on, end-to-end, and built for the real world.

GENERATIVE AI ENGINEER

Skills Required

Advanced GenAI, LLMs (GPT-4, Cohere), Hugging Face, Diffusion Models, GANs, VAEs, Cloud ML Platforms (AWS SageMaker, Azure ML)

₹12 - 30 LPA STARTING SALARY

AI ENGINEER

Skills Required

Python, Machine Learning basics, TensorFlow/PyTorch, LLM fine-tuning, Prompt Engineering, LangChain, AutoGen, CrewAI

₹12 - 25 LPA STARTING SALARY

MACHINE LEARNING ENGINEER

Skills Required

Python, Basic TensorFlow/PyTorch, Model Training, Data Preprocessing, Feature Engineering, Model Evaluation

₹12 - 28 LPA STARTING SALARY

DATA SCIENTIST

Skills Required

Statistical Analysis, Python, R, SQL, Data Visualization, Machine Learning fundamentals

₹12 - 22 LPA STARTING SALARY

AIOPS ENGINEER

Skills Required

Basic Docker, Simple CI/CD, Model Deployment, Kubernetes basics, Prometheus, Grafana, Cloud platforms

₹15 - 35 LPA STARTING SALARY

DATA ANALYST

Skills Required

SQL, Excel, Basic Python, Tableau/PowerBI, Statistical Analysis, Data Visualization

₹12 - 18 LPA STARTING SALARY

AI PRODUCT ANALYST

Skills Required

Comprehensive AI/ML understanding, Business Impact Analysis, Data-driven insights, Product Management basics

₹15 - 30 LPA STARTING SALARY

CLOUD AI ENGINEER

Skills Required

Basic AWS/Azure, Simple Deployments, Cloud Security, Serverless AI, Auto-scaling ML Infrastructure

₹18 - 35 LPA STARTING SALARY

*Disclaimer: Job outcomes vary based on experience, role, employer, location, and market conditions.

Curriculum

Module 1: Foundations of AI, ML, and Generative AI

Topics

- Understanding Supervised Machine Learning, Unsupervised Machine Learning, and Reinforcement Learning
- Data Manipulation for Machine Learning
- Exploratory Data Analysis for Machine Learning
- Essential Mathematics for Machine Learning and AIDefinition of Generative AI
- Generative AI Applications
- Definition of LLMs

Module 2: Deep dive into Machine Learning Algorithms

Topics

- Regression Supervised Machine Learning
  Linear Regression
  Decision Tree Regressor
  Random Forest Regressor
  Overfitting, Underfitting, and Regularization
- Classification Supervised Machine Learning
  Logistic Regression
  Decision Tree Classifier
  Random Forest Classifier
- Unsupervised Machine Learning
  K-Means Clustering
  Hierarchical Clustering
  Dimensionality Reduction
  Anomaly detection
- Understanding Reinforcement learning: Basic concepts (Action, reward, process), MPD, RL, DRL, DQN, etc.
- MLOps for Machine Learning model deployment

Module 3: Understanding Deep Learning

Topics

- Exploring Deep Learning
  Understanding Neural Networks with TensorFlow
- Deep dive into Neural Networks with TensorFlow
- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Object detection using the YOLO framework
- Transfer Learning in CNN
- Recurrent Neural Networks (RNN) and LSTMs

Module 4: Exploring NLP

Topics

- Architectures of RNN - One to One, Many to Many, Many to One, etc
- Understanding Text for NLP
  Tokenization, Stemming and Lemmatization, Stopwords, and Keywords in NLP
- Text Vectorization using TF-IDF, Bag of Words, and Word 2 Vec
- Use cases across domains in NLP

Module 5: Engaging with Generative AI

Topics

- How Generative AI works
- Text generation, Image generation
- Audio generation and video generation
- Using Hugging face to access models for text generation, image generation
- GANs
  Types of GANs - ProGAN, SRGan, CycleGAN
- Auto Encoders, Variational Auto Encoders (VAEs), Diffusion Models

Module 6: Understanding LLMs

Topics

- Definition of LLMs
  LLM Use Cases
- Prompt Tuning
- Attention Mechanism
- Transformer Model and Architecture
- Encoder-Decoder arrangements
- Train and generate text using
- Encoder-decoder Architecture
- BERT for Transfer learning
- Leveraging multiple pretrained LLMs from Hugging Face
- Fine-tuning LLMs

Module 7: GenAI Application Development Prompting Techniques

Topics

- Introduction to Prompt Engineering
- Successful and Unsuccessful prompts
- Types of Prompting
- Introduction to Open AI, GPT , Open AI Playground
- Cost & latency considerations when calling APIs (OpenAI, Azure, AWS).
- Multimodal prompting for GPT 4
- Image generation using Open AI DALLE 3
- Prompt evaluation
- Implementing Agents and Chains
- Implementing zero-shot-react, conversational-react agents with LangChain
- Open AI Function (Tool Integration)
- Testing various LLMs with Prompt Engineering

Module 8: Synthetic Data and Datasets for LLMs

Topics

- Introduction to Synthetic Data
- Generating Synthetic Data
- Synthetic Data for LLMs
- Real-world Applications and Use Cases
- Hands-on generating and using Synthetic Data

Module 9: AI Embeddings & Retrieval

Topics

- Understanding AI Embeddings
- Advanced Retrieval Techniques
- Hugging Face Embeddings
- Vector Databases in AI
- CRUD operations with Vector Databases
- RAG - Retreival Augmented Generation
- RAG solutions using Open AI models and Hugging face models
- Ethical Considerations in AI Embeddings
- Navigating AI Hallucinations, Drift, and Bias
- Embeddings in Real-world Applications
- Embeddings Optimization and Fine-tuning
- Embeddings Security and Privacy
- LLM Ops and model deployment best practices

Module 10: Understanding Agentic AI

Topics

- Agents, Agentic AI, and Multi-Agent Systems
- Agent Definition & Autonomy
  Simple vs. Knowledge-Based Agents
- Reflex vs. Goal-Driven Agents
  Microsoft AutoGen
- Agent Architecture (Perception, Decision, Action)
- Integrating Knowledge Bases (RAG, Domain Data)
- Measuring Performance (Success Rate, Resource Usage)
- Hierarchical Agent Planning
- Multi-Step Reasoning with LLM
- Memory & Long-Term Context
- Integrating Retrieval Augmentation in Agent Workflows
- Domain-Specific Knowledge & Dynamic Prompting

Module 11: Exploring Cloud Ecosystems

Topics

- Cloud Ecosystems
- Introduction to Cloud Ecosystem
  Definitions
- Cloud characteristics
- Deployment models
- Leading Service providers (AWS, Google, Azure, etc.)
- Comparing AWS, Azure, and GCP core services for compute, storage, and AI/ML.
- Data Centres and their components
- Service (SaaS, IaaS, PaaS)
  Issues & Challenges

Module 12: Advancement in Hypervisors

Topics

- Understanding hypervisors
  Reference model
- Virtualisation characteristics
- Principles of hypervisor design Interfaces
- Types of hypervisors (type-1 and type-2)
- Differences between Type-1 and Type-2 hypervisors.
- Design methods of hypervisors (full virtualization, para virtualization, and hardware-assisted virtualization)
- Memory Virtualisation
- I/O virtualisation
- OS virtualization
- Comparative Analysis of hypervisors
- Understanding performance, requirements, and bottleneck

Module 13: Container Orchestration

Topics

- Understanding LLM Deployment Architectures
- Containerizing LLM Inference Services (e.g., using FastAPI + Docker)
- Managing GPU Workloads in Kubernetes
- Scaling LLM APIs with Kubernetes and Istio
- Optimizing Latency and Throughput for LLM Containers
- Secure Access and Rate Limiting for AI APIs
- CI/CD for LLM-Powered Microservices
- Monitoring and Logging for LLM Containers
- Model Versioning and Rollbacks
- Cost Optimization Strategies for LLM Inference in Production

Module 14: Cloud Security & Resilience

Topics

- Infrastructure security: Network-level security, Host-level security, Application-level security
- Data security and storage: Data privacy and security issues Jurisdictional issues raised by data location
- Identity and access management
- Access control
- IAM, Key Management Services, and zero-trust architecture
- Trust, reputation, risk
  Authentication in cloud computing
- Client access in the cloud
- Cloud contracting model
- Commercial and business considerations

Module 15: Understanding Cloud in context of Gen AI and LLMs

Topics

- EC2 Deep Dive and AMIs, EBS vs S3 vs EFS – Storage Solutions
- Load Balancing and Auto Scaling Basics
- Intro to Serverless: AWS Lambda
- Using AWS Bedrock for GenAI (including foundation models)
- Deploying Open-Source LLMs on EC2/EKS
- Fine-Tuning and Inference
- Pipelines on Cloud
- Cost & Performance
- Considerations for LLM Workloads
- SageMaker Pipelines for Model
- Training and Inference
- Model Versioning, A/B Testing, and Rollbacks
- Security and Compliance for GenAI in Production

Program –

Outcomes

By the end of this program, learners will be equipped with the capabilities to operationalize the product for real-world impact.

Design and implement end-to-end AI and ML workflows

Including data preparation, supervised & unsupervised model training, and evaluation metrics using real-world datasets.

Demonstrate expertise in core ML, deep learning, and NLP

Master neural networks, CNNs, RNNs, and transformers for predictive modeling and text-based AI systems.

Apply advanced Generative AI methods

Use GANs, VAEs, and diffusion models to generate text, images, & data for both synthetic experimentation and creative automation.

Build, fine-tune, and deploy Large Language Models (LLMs)

Use frameworks like Hugging Face, LangChain, & CrewAI to create domain-specific GenAI applications through effective prompt engineering

Develop and manage AI-powered agents and multi-agent systems

Leverage Agentic AI stacks using orchestration tools and retrieval-augmented generation techniques to automate complex workflows.

Deploy AI applications on the cloud

Containerize and scale AI systems using Docker, Kubernetes, AWS, and GCP while maintaining compliance, observability, and uptime.

Operationalize AI through AIOps and MLOps

Ensure retraining, rollback, and versioning of models using pipelines that support continuous learning and monitoring.

Address AI safety, ethics, and governance

Understand the legal and ethical implications of AI and build applications that are secure, responsible, and regulation-ready.

Translate AI into a measurable business impact

Bridge the gap between models & outcomes with solutions that directly improve KPIs across domains like finance, marketing, operations, & product.

END-TO-END PROJECT-BASED LEARNING

Work on real-world projects aligned to enterprise-grade use-cases:

- GenAI Document Processors
- AI-Enabled Workflow Automation
- LangChain-based Customer Assistants
- Autonomous Agent Workflows
- Full-stack Cloud AI Deployments